DUY TÂN UNIVERSITY

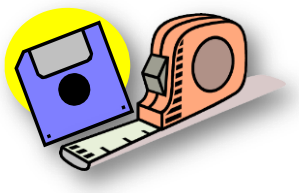
CMU-CS 462: Software Measurement and Analysis

Spring 2025-2026

Nguyen Duc Man
E: mannd@duytan.edu.vn
T: 0904235945

1

Measuring Internal Product Attributes: Software Size (cont'd)



DUY TÂN UNIVERSITY

1. (0.2 points)

In size metrics, metrics are developed based on the _____

a. number of Functions
b. number of user inputs
c. number of lines of code
d. amount of memory usage

1. (0.2 points)

Which of the following are information domains required for determining function point in FPA (nhiều phương án)?

a. Number of user Input
b. Number of user Inquiries
c. Number of external Interfaces
d. Number of errors

3

DUY TÂN UNIVERSITY

1. (0.2 points)

How to measure the defect density of a system (more than one)?

a. defects per KLOC

b. defects per function point

c. total number of defects

d. None of above

1. (0.2 points)

Which statement is not a measurement of Length in LOC?

a. Count of physical lines including blank lines.

b. Count of all Operators

c. Count of all lines except blank lines and comments.

d. Count of all statements except comments

4

DUY TÂN UNIVERSITY

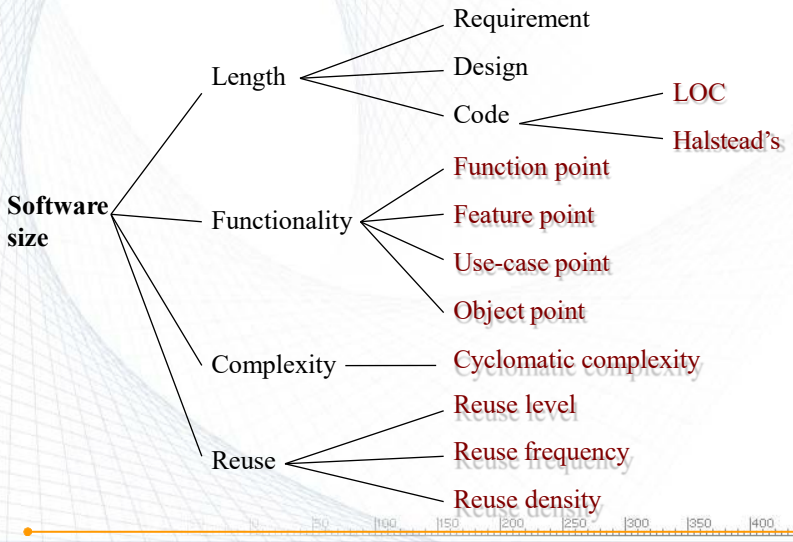
Contents

1. Software size
2. Size: Length (code, specification, design)
3. Size: Reuse
4. **Size: Functionality (function point, feature point, object point, use-case point)**
5. **Size: Complexity**

5

DUY TÂN UNIVERSITY

Software Size Metrics: Summary



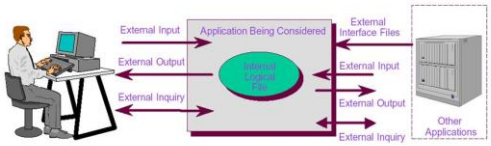
6

DUY TÂN UNIVERSITY

Function Points (IFPUG)

Example 1

Function Points (FPA
Software Model



To identify : EI-EO-EQ & ILF-EIF

7

DUY TÂN UNIVERSITY

FP Example /1a

Specification: Spell Checker

- Checks all words in a document by comparing them to a list of words in the internal dictionary and an optional user-defined dictionary
- After processing the document sends a report on all misspelled words to standard output
- On request from user shows number of words processed on standard output
- On request from user shows number of spelling errors detected on standard output
- Requests can be issued at any point in time while processing the document file

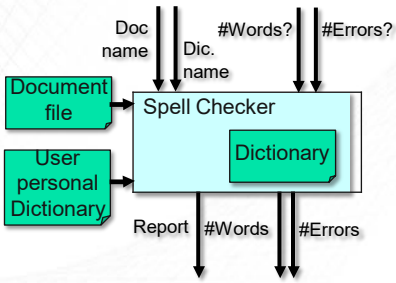
8

DUY TÂN UNIVERSITY

FP Example /1b

Spell Checker

- Convert spec to a diagram



Specification:

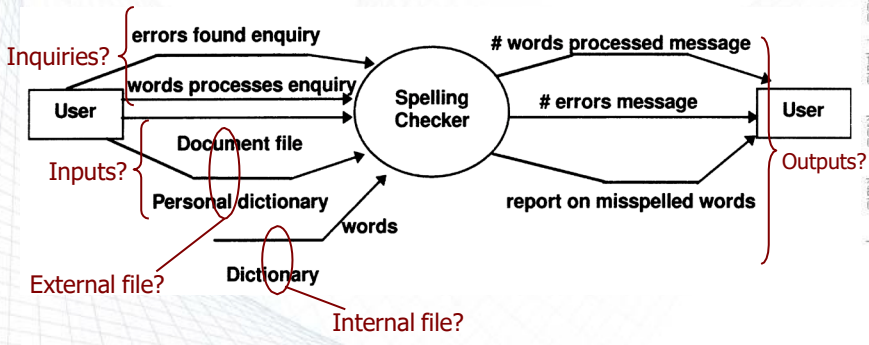
1. Checks all words in a document by comparing them to a list of words in the internal dictionary and an optional user-defined dictionary
2. After processing the document sends a report on all misspelled words to standard output
3. On request from user shows number of words processed on standard output
4. On request from user shows number of spelling errors detected on standard output
5. Requests can be issued at any point in time while processing the document file

DUY TÂN UNIVERSITY

FP Example /1c

Spell Checker

- Convert spec to a diagram

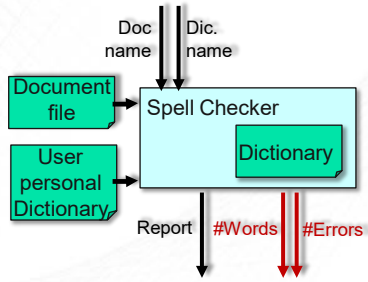


DUY TÂN UNIVERSITY

FP Example /1d

Solution A:

- EI: Doc. name + User Dic. name → 2
- EO: Report + #Words + #Errors → 3
- EQ: --
- EIF: Document + User Dictionary → 2
- ILF: Dictionary → 1



EI EO EQ EIF ILF

DUY TÂN UNIVERSITY

FP Example /1e

Element	Weighting Factor		
	Simple	Average	Complex
External inputs (*EI*)	3	4	6
External outputs (*EO*)	4	5	7
External inquiries (*EQ*)	3	4	6
External interface files (*EIF*)	5	7	10
Internal logical files (*ILF*)	7	10	15

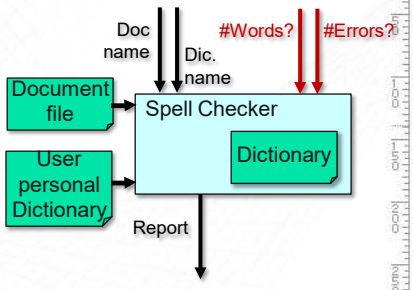
- $N_{EI}=2$ (number of EI): document name, user-defined dictionary name
- $N_{EO}=3$ (number of EO): misspelled word report, number-of-words-processed message, number-of-errors-so-far message
- $N_{EQ}=0$ (number of EQ): --
- $N_{EIF}=2$ (number of EIF): document file, personal dictionary
- $N_{ILF}=1$ (number of ILF): dictionary

DUY TÂN UNIVERSITY

FP Example /1f

Solution B:

- EI: Doc. name + Dic. name → 2
- EO: Report → 1
- EQ: #Words? + #Errors? → 2
- EIF: Document + User Dictionary → 2
- ILF: Dictionary → 1



EI EO EQ EIF ILF

DUY TÂN UNIVERSITY

FP Example /1g

Element	Weighting Factor		
	Simple	Average	Complex
External inputs (*EI*)	3	4	6
External outputs (*EO*)	4	5	7
External inquiries (*EQ*)	3	4	6
External interface files (*EIF*)	5	7	10
Internal logical files (*ILF*)	7	10	15

- $N_{EI}=2$ (number of EI): document name, user-defined dictionary name
- $N_{EO}=1$ (number of EO): misspelled word report
- $N_{EQ}=2$ (number of EQ): number-of-words-processed message, number-of-errors-so-far message
- $N_{EIF}=2$ (number of EIF): document file, personal dictionary
- $N_{ILF}=1$ (number of ILF): dictionary

DUY TÂN UNIVERSITY

FP Example /1h

- Suppose that
  $F_1=F_2=F_6=F_7=F_8=F_{14}= 3$;
  $F_4=F_{10}=5$
  and the rest are zero.

$$VAF = 0.65 + 0.01\sum_{j=1}^{14} F_j$$

Solution A:
Solution B:

DUY TÂN UNIVERSITY

FP Example /1i –Discussion

Which solution is correct?

- Solution A?
- Solution B?
- None of them? (cf. Textbook)

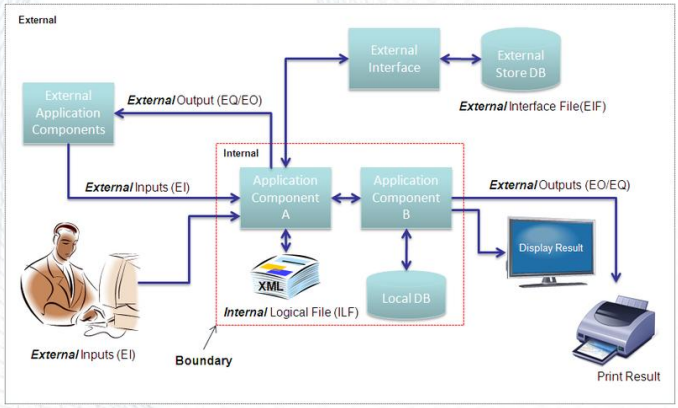
Answer: Solution A is correct!

Why? The difference is because of correct interpretation of EO and EQ for the requirement #5

Wrong Calculation in Fenton's book

DUY TÂN UNIVERSITY

Function Points (IFPUG)



Example 2

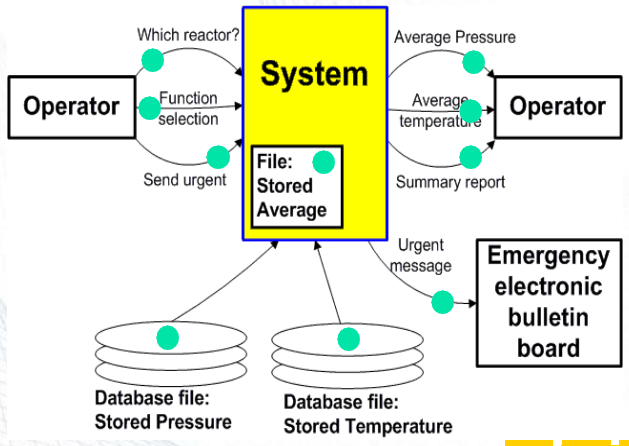
DUY TÂN UNIVERSITY

FP: Example /2a

- Consider the following system that processes various commands from the operator of a chemical plant. Various interactions of the system with the operator and internal and external files are shown.
- The system consults two databases for stored pressure and temperature readings. In the case of emergency, the system asks the user whether to send the results to an electronic emergency bulletin board or not.
- Calculate the unadjusted function point count (UFC) for this system.

DUY TÂN UNIVERSITY

FP: Example /2b

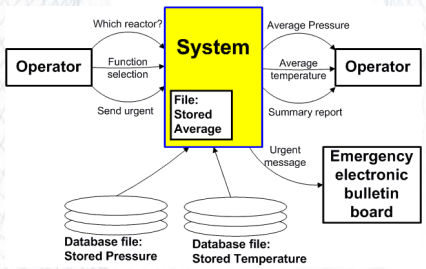


EI EO EQ EIF ILF

19

DUY TÂN UNIVERSITY

FP: Example /2c

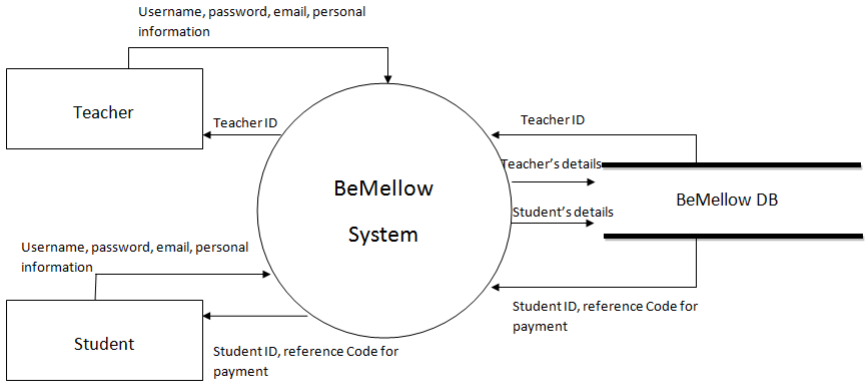


- N_{EI} number of external inputs **2**
- N_{EO} number of external outputs **3**
- N_{EQ} number of external inquiries **1**
- N_{EIF} number of external interface files **2**
- N_{ILF} number of internal logical files **1**

$$UFC = 4N_{EI} + 5N_{EO} + 4N_{EQ} + 7N_{EIF} + 10N_{ILF}$$

UFC = 4x2 + 5x3 + 4x1 + 7x2 + 10x1 = 51

20



21

Review: Software Size

- Size measurement must reflect *effort*, *cost* and *productivity*.
- Defining software size in terms of length, functionality, and complexity, each capturing a key aspect of software size.
- Basic measure for length is LOC.
- Basic measure for functionality is Function Point (FP).
- FP is computed in two steps:
  1) Calculating an ***Unadjusted Function point Count (UFC)***.
  2) Multiplying the UFC by a ***Value Adjustment Factor (VAF)***.

  The final (adjusted) Function Point is:

  $$FP = UFC \times VAF$$
- FP variations: feature point, object point, use-case point.

22

DUY TÂN UNIVERSITY

FP vs. LOC /1

- A number of studies have attempted to relate LOC and FP metrics (Jones, 1996).
- The average number of source code statements per function point has been derived from case studies for numerous programming languages.
- Languages have been classified into different levels according to the relationship between LOC and FP.

DUY TÂN UNIVERSITY

FP vs. LOC /2

Language	Typical SLOC per FP
1GL Default Language	320
2GL Default Language	107
3GL Default Language	80
4GL Default Language	20
Code generators	15
Assembler	320
C	148
Basic	107
Pascal	90

DUY TÂN UNIVERSITY

FP vs. LOC /3

Language	Typical SLOC per FP
C#	59
C++	60
PL/SQL	46
Java 2	60
Visual Basic	50
Delphi	18
HTML 4	14
SQL	13
Excel	47

Newer Data!
http://www.qsm.com/FPGearing.html

25

DUY TÂN UNIVERSITY

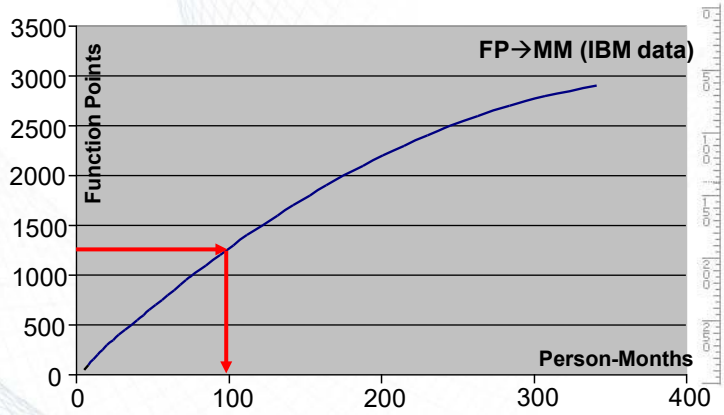
```
01 DATA SEGMENT
02       NUM1 DB 9H
03       NUM2 DB 7H
04       RESULT DB ?
05 ENDS
06
07 CODE SEGMENT
08     ASSUME DS:DATA CS:CODE
09 START:
10       MOV AX,DATA
11       MOV DS,AX
12
13       MOV AL,NUM1
14       ADD AL,NUM2
15
16       MOV RESULT,AL
17
18       MOV AH,4CH
19       INT 21H
20 ENDS
21 END START
22
```


26

DUY TÂN UNIVERSITY

Effort Estimation with FP

Effort estimation based on organization-specific data from past projects.



DUY TÂN UNIVERSITY

FP: Advantages –Summary

- Can be counted before design or code documents exist
- Can be used for estimating project cost, effort, schedule early in the project life-cycle
- Helps with contract negotiations
- Is standardized (though several competing standards exist)

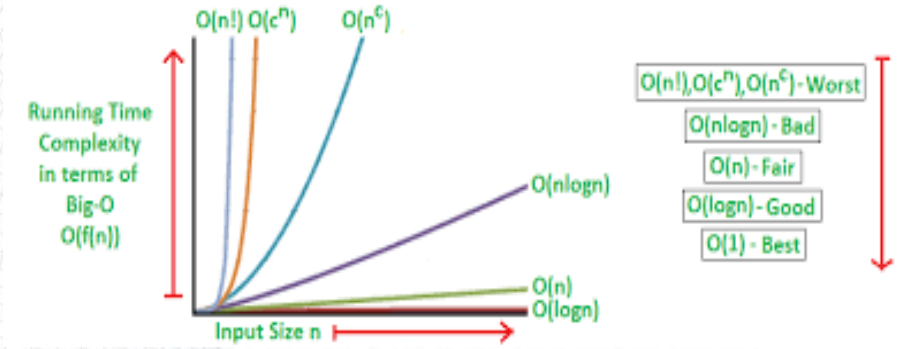
DUY TÂN UNIVERSITY

FP: Critics

- FP is a subjective measure: affected by the selection of weights by external users.
- Function point calculation requires a full software system specification. It is therefore difficult to use function points very early in the software development lifecycle.
- Physical meaning of the basic unit of FP is unclear.
- Unable to account for new versions of I/O, such as data streams, intelligent message passing, etc.
- Not suitable for "complex" software, e.g., real-time and embedded applications.

DUY TÂN UNIVERSITY

Feature Point



DUY TÂN UNIVERSITY

Feature Point /1

- Function points were originally designed to be applied to business information systems. Extensions have been suggested called ***feature points*** which may enable this measure to be applied to other software engineering applications.
- Feature points accommodate applications in which the "algorithmic complexity" is high such as real-time, process control, and embedded software applications.
- For conventional software and information systems functions and feature points produce similar results. For complex systems, feature points often produce counts about %20~%35 higher than function points.

DUY TÂN UNIVERSITY

Feature Point /2

- Feature points are calculated the same way as FPs with the additions of ***algorithms*** as an additional software characteristic.
- Counts are made for the five FP categories, i.e., number of external inputs, external outputs, inquiries, internal files, external interfaces, plus:
  - **Algorithm (N_a):** A bounded computational problem such as inverting a matrix, decoding a bit string, or handling an interrupt.

DUY TÂN UNIVERSITY

Feature Point /3

- Feature points are calculated using:
  - Number of external inputs ×4
  - Number of external outputs ×5
  - Number of external inquiries ×4
  - Number of external interface files ×7 ←
  - Number of internal files ×10
  - Algorithms ×3 ←

$$UFeC = 4N_{EI} + 5N_{EO} + 4N_{EQ} + 7N_{EIF} + 10N_{ILF} + 3N_{A}$$

- The *UFeC* used in the function point calculation to calculate the feature points.

DUY TÂN UNIVERSITY

Object Point /1

- Object points are used as an initial measure for size way early in the development cycle, during feasibility studies.
- An initial size measure is determined by counting the number of ***screens***, ***reports***, and third-generation ***components*** that will be used in the application.
- Each object is classified as ***simple***, ***medium***, or ***difficult***.

DUY TÂN UNIVERSITY

Object Point /2

- Object point complexity levels for screens.

Number of views contained	Number and source of data tables		
	Total <4 <2 servers <2 clients	Total <8 2-3 servers 3-5 clients	Total 8+ >3 servers >5 clients
< 3	Simple	Simple	Medium
3 - 7	Simple	Medium	Difficult
8+	Medium	Difficult	Difficult

DUY TÂN UNIVERSITY

Object Point /3

- Object point complexity levels for reports.

Number of views contained	Number and source of data tables		
	Total <4 <2 servers <2 clients	Total <8 2-3 servers 3-5 clients	Total 8+ >3 servers >5 clients
0 - 1	Simple	Simple	Medium
2 - 3	Simple	Medium	Difficult
4+	Medium	Difficult	Difficult

DUY TÂN UNIVERSITY

Object Point /3

- The number in each cell is then weighted and summed to get the object point.
- The weights represent the relative effort required to implement an instance of that complexity level.
- **Complexity level for object point:**

Object Type	Simple	Medium	Difficult
Screen	1	2	3
Report	2	5	8
3GL component	-	-	10

New object points = (object points) ×(100 – r) / 100

Assuming that *% r* of the objects will be reused from previous projects.

DUY TÂN UNIVERSITY

How to Calculate OP?

Work in pairs (thảo luận theo nhóm 2)!

DUY TÂN UNIVERSITY

Effort Estimation with OP

- Formula:

  Effort [Person-Month] = OP / PROD

- How to measure PROD?

Developer Experience/Skills	Very Low	Low	Average (Nominal)	High	Very High
PROD	4	7	13	25	50

- Example:

  Effort [Person-Month] = 13 / 13 = 1

  Assuming average experience and 0% reuse.

DUY TÂN UNIVERSITY

Example: OP

- Suppose that you have
  - 4 screens, 2 of them simple (weight 1) and 2 of them medium (weight 2)
  - 3 reports, 2 of them simple (weight 2) and 1 medium (weight 5),

  then the total number of OP is:

  $OP = 2\times1 + 1\times2 + 2\times2 + 1\times5 = 13$
- If you have any acquired component, give it the weight 10 and add it to the OP.
- If you have any part of your system reused from a previous generation, define a percentage of reuse (say 10%) and then adjust the value of OP accordingly.

  $OPnew = 13 \times (100\text{-}10)/100 = 11.7$

DUY TÂN UNIVERSITY

Example 2

- Determine object points for a smart vending machine assuming that the system is average size (total number of clients-servers is less than 8) using the following data:

Screens	***Views (Data Items)***
User Screens	
S1. Buying Screen	Amount of coins to insert, selection
Administrator Screens	
S2. Inventory Update	Inventory input update (20 data items)
S3. Change Update	Change infrared input update (4 data items)

DUY TÂN UNIVERSITY

Example 2 (cont'd)

Reports	***Views (Data Items)***
User Reports	
R1. Display money entered	Amount of money entered
R2. Not enough money entered	Insufficient funds
R3. Returning change	Amount of change
R4. Out of stack message	Out of stock
R5. No change message	No change
R6. Selection Approved	Release selection signal
Administrator Reports	
R7. Machine Report	Inventory (20 data items), change information (4 data items)

DUY TÂN UNIVERSITY

Example 2 (cont'd)

Simple	*Medium*	*Difficult*
S1	*S3*	*S2*
R1 R2 R3 R4 R5 R6	*None*	*R7*

OP = (1×1) + (1×2) + (1×3) + (2×6) + (5×0) +(8×1)
OP = 26

Effort = 26/13 = 2 person month

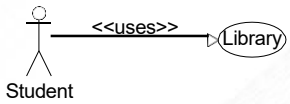
DUY TÂN UNIVERSITY

Use-Case Point

DUY TÂN UNIVERSITY

Use-Case Point /1

- *Function Point* is a method to measure software size from a *requirements* perspective.
- *Use-Case* is a method to develop *requirements*. Use Cases are used to validate a proposed design and to ensure it meets all requirements.



Question: How to use Use-Cases to measure function point and vice-versa?

DUY TÂN UNIVERSITY

Use-Case Point /2

Question: How to use Use-Cases to measure function point and vice-versa?

Two methods:

1. Identify and weight ***actors*** and ***use-cases***
2. Count the inputs, outputs, files and data inquiries from use-cases (using the ***use-case definition*** and ***activity diagram***).

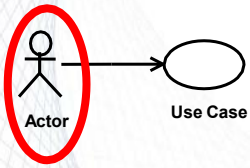
DUY TÂN UNIVERSITY

Use Case Point /3

- Use Case Point Estimation has its origin in the work by Gustav Karner (1993)
- Process:
  1. Identify and weight Actors (→ UAW)
  2. Identify and weight Use Cases (→ UUCW)
  3. Calculate Unadjusted Use Case Points (→ UUCP = UAW + UUCW)
  4. Calculate Value Adjustment Factor (→ VAF)
  5. Calculate (Adjusted) Use Case Points
     → UCP = UUCP * VAF

DUY TÂN UNIVERSITY

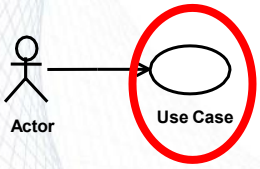
Use Case Point: Actors



- For each Actor, determine whether the actor is simple, average or complex.
- A ***simple actor*** represents another system with a defined Application Programming Interface (API). [weight: 1]
- An ***average actor*** is either another system that interacts through a protocol (HTTP, FTP, user-defined, etc.), a data store (files, DBs, etc.) or a person interacting through a text-based interface. [weight: 2]
- A ***complex actor*** is a person interacting through a graphical user interface (GUI). [weight: 3]

DUY TÂN UNIVERSITY

Use Case Point: UseCases



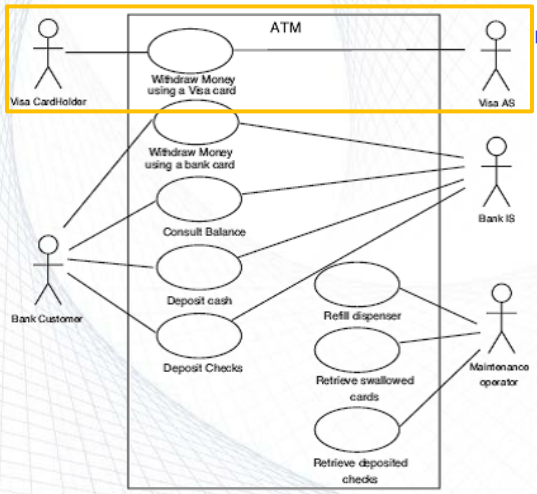
- For each Use Case, determine whether it is simple, average or complex.
  - A **simple use case** contains up to 3 transactions. [weight: 5]
  - An **average use case** contains 4-7 transactions. [weight: 10]
  - A **complex use case** contains more than 7 transactions. [weight: 15]
- ***Note:*** The original Use Case Point method suggested to count only transactions of the main (success) scenarios. Some researchers propose to include in the counting also alternate (extensions) and exception (error) transaction sequences.

DUY TÂN UNIVERSITY

Tên use case: Đăng Nhập	
Mô tả sơ lược: Người dùng muốn truy cập vào hệ thống để sử dụng các tính năng liên quan đến đăng ký học phần.	
- Actor chính: Sinh viên/Giảng viên	
- Actor phụ: không	
- Pre-condition: Người dùng đã có tài khoản trong hệ thống.	
- Post-condition: Người dùng đã đăng nhập thành công và có thể sử dụng các tính năng của hệ thống.	
Luồng sự kiện chính (main flow):	
Actor	**System**
1. Người dùng truy cập trang đăng nhập của hệ thống.	2. Hệ thống hiển thị trang đăng nhập
3. Người dùng nhập tên đăng nhập và mật khẩu của mình.	4. Hệ thống kiểm tra thông tin đăng nhập
	5. Hệ thống thông báo đăng nhập thành công
	6. Hệ thống hiển thị giao diện chính
Luồng sự kiện thay thế (alternate flow):	
	4.1 Hệ thống thông báo thông tin đăng nhập sai
4.2 người dùng xác nhận	4.3 Hệ thống quay lại bước 2

DUY TÂN UNIVERSITY

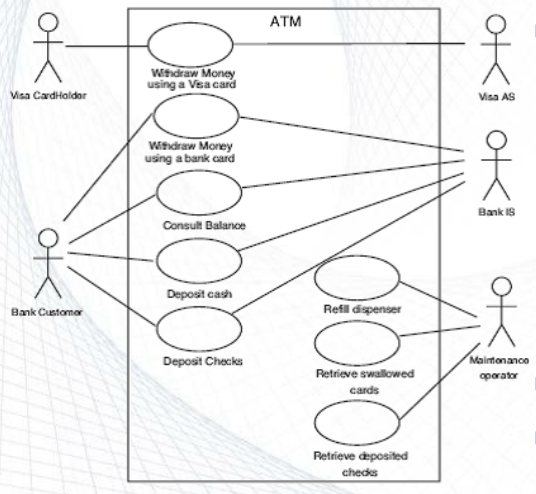
Example: ATM



- Actors:
  - Visa Cardholder: complex (interacts via GUI)
  - Visa AS: average (interacts via communication protocol)

DUY TÂN UNIVERSITY

Example: ATM



- Use Case:
  - "Withdraw money using a Visa card" complex (more than 7 transactions, even without counting extensions or exceptions)
- UAW = 3 + 2 = 5
- UUCW = 15

DUY TÂN UNIVERSITY

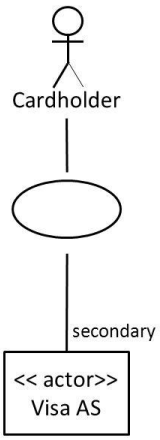
ATM: Use Case Body: Withdraw Money (Visa)

ENGR 110 10:15

Withdraw Money Using Visa Card (Actors: Cardholder, Visa AS)	
1. User inserts card	2. Request PIN
3. User enters PIN	4. Request authorisation
5. Visa AS confirms	6. Show options
7. User selects Withdraw	8. Ask for desired amount
9. User enters amount	10. ATM requests limit
11. Visa AS reports limit	12. ATM checks if under limit
	13. ATM returns card
14. User takes card	15. ATM issues banknotes
16. User takes money	

User Intention

System responsibility

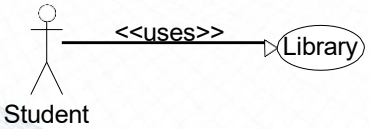


53

DUY TÂN UNIVERSITY

Use-Case Point /2

- We must count the inputs, outputs, files and data inquiries from use-cases.
- Function points become evident using the ***use-case definition*** and ***activity diagram*** for the use-case. Each step within the activity diagram can be a transaction (inputs, outputs, files and data inquiries).



54

DUY TÂN UNIVERSITY

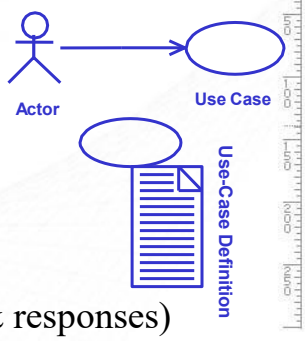
Examining Use Case Scenario

- Use Case might be further refined via Activity and Interaction Diagrams
- Examining Use Case definition and Activity and/or Interaction diagrams can also lead to FP count.
- How?

DUY TÂN UNIVERSITY

Use-Case Definition

- A typical ***use-case definition*** document consists of:
  - Use Case name
  - Actor name
  - Objective
  - Preconditions
  - Results (Post-conditions)
  - Detailed description (actions & responses)
  - Exceptions and alternative courses



Use-Case Definition Worksheet

Use Case Definition

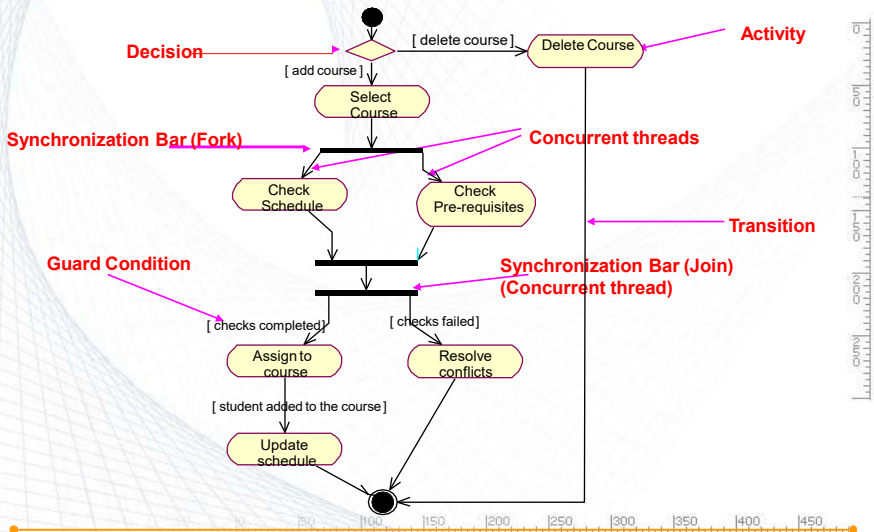
Last Updated by: Last Updated On:
Use Case Name: UC ID:
Actor:
Objective:
Preconditions:
Results (Postconditions):
Detailed Description

Action	Model (System) Response
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.

Exceptions:
Alternative Courses:
Original Author: Original Date:

DUY TÂN UNIVERSITY

Example: Activity Diagram



DUY TÂN UNIVERSITY

Example 1: Use-Case Scenario

Use-Case: Library

Scenario:

1. The user may enter a book's ISBN number, student ID, or student name
2. The user will press "Find"
3. If the user enters a book ISBN number
   - The system will display information related to that book and write the results to a file
4. If the user entered a student name or student ID
   - The system will return a list of all books on the waiting list for that student and write the results to a file
   - The user can select one book from the list
   - The system will search the database by ISBN number
   - The system will display information related to that book and available date and will write the results to a file

Together they will qualify as 1 output

DUY TÂN UNIVERSITY

Example 1: Use-Case Scenario

Use-Case: Library

Scenario:

But these are not independent from the "Find" input, so they are actually data attributes for "Find"

1. The user may enter a book's ISBN number, student ID, or student name ***(input ×3)***
2. The user will press "Find" ***(input ×1)***
3. If the user enters a book ISBN number ***(input ×1)***
   - The system will display information related to that book and write the results to a file ***(output ×1) (internal file ×1)***
4. If the user entered a student name or student ID ***(input ×2)***
   - The system will return a list of all books on the waiting list for that student and write the results to a file ***(output ×1) (internal file ×1)***
   - The user can select one book from the list ***(external inquiry ×1)***
   - The system will search the database by ISBN number ***(external file ×1)***
   - The system will display information related to that book and available date and will write the results to a file ***(output ×1) (internal file ×1)***

Together they will qualify as 1 output

DUY TÂN UNIVERSITY

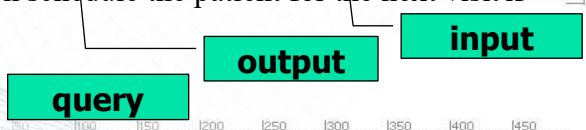
Example 2 (work in team 3)

- Dr. X, a recent graduate from a medical university, is starting her medical practice in a small town. She is planning to hire a receptionist. She approaches software company Y to build a software system to manage the patients' appointments. The following is her problem description (next two slides).
- Suppose that you are the analyst in charge of estimating the cost of this system and you base your estimation of the function point (FP) count for this system.
- Calculate the unadjusted function point count (UFC) for the system assuming the average weight for all entries.

DUY TÂN UNIVERSITY

Example 2 (cont'd)

- When a patient calls for an appointment, the receptionist will ask the **patient's name** or **patient's ID number** and will **check the calendar** and will try to **schedule** the patient as early as possible to fill in vacancies. If the patient is happy with the proposed **appointment**, the receptionist will enter the appointment with the **patient name** and **purpose** of appointment. The system will verify the **patient name** and supply **supporting details** from the **patient records**, including the **patient's ID number**. After each appointment the Dr. X will **mark** the appointment as completed, **add comments**, and then schedule the patient for the next visit if appropriate.



DUY TÂN UNIVERSITY

Example 2 (cont'd)

Note that we have to again ask ourselves whether these qualify for EI, EO, EQ, EIF and ILF.

Type	Description	No.	Weight	Weighted value
Inputs (EI)	Patient name, Patient ID number, Appointment completed, Appointment purpose, Cancel appointment	5	4	20
Outputs (EO)	Comments, Calendar, Supporting details, Appointment Information, Notification List, Daily schedule, Weekly schedule,	7	5	35
Queries (EQ)	Check calendar, Query by name, Query by ID, Query by date, Verify patient, Available Appointment	6	4	24
Internal files (ILF)	Patients' data record	1	10	10
External files (EIF)	--	0	7	0
		Total UFC		89

65

DUY TÂN UNIVERSITY

Recent Approaches

- Recent approaches to size measurement and estimation include:
  - Decomposition
  - Expert opinion
  - Analogy
  - Class-code expansion

66

DUY TÂN UNIVERSITY

4. Class–Code Expansion

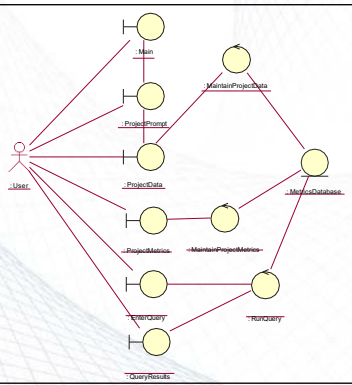
- In the design phase, boundary classes are usually expanded by the ratio of 1 to 2 and control classes are expanded by the ratio of 2 to 3, respectively. The entity classes are usually expanded to a subsystem having 4 to 8 design classes.
- An average design class in Java has 20 methods, each method having 10 to 20 lines of code.

Analysis Classes Description	
	Boundary class
	Control class
	Entity class

DUY TÂN UNIVERSITY

Example

- Using class-code expansion

Model	*Description*	
		Boundary class
		Control class
		Entity class

DUY TÂN UNIVERSITY

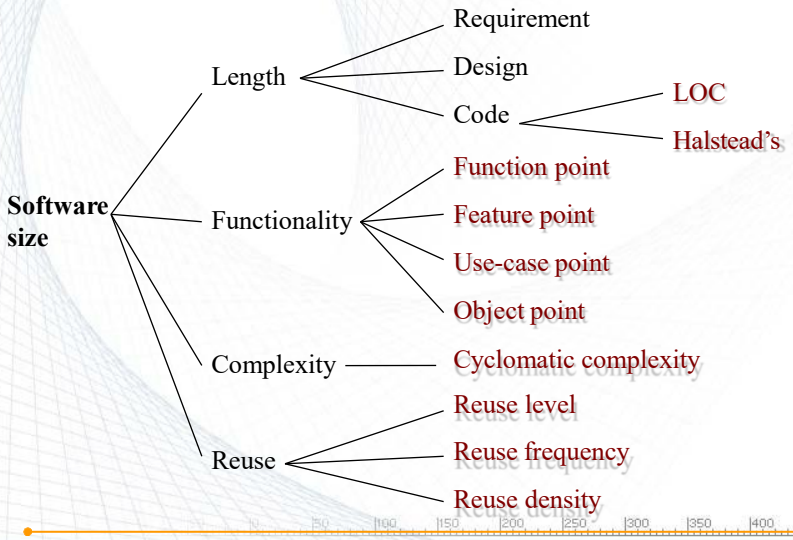
Example (cont'd)

- Calculate total software size in LOC, assuming total size is 1.5 × methods LOC
- Calculate total development cost.

Analysis classes	*No.*	*Total classes*	*Total Operations*	*Total LOC*
Boundary class	**6**	**Min 16**	**Min 320**	**1.5 × 320 × 10 = 4,800 (minimum)**
min (× 1)	**6**	**Max 29**	**Max 580**	**1.5 × 320 × 20 = 9,600**
max (× 2)	**12**			**1.5 × 580 × 10 = 8,700**
Control class	**3**			**1.5 × 580 × 20 = 17,400 (maximum)**
min (× 2)	**6**			
max (× 3)	**9**			
Entity class	**1**			
min (× 4)	**4**			
max (× 8)	**8**			

DUY TÂN UNIVERSITY

Software Size Metrics: Summary



DUY TÂN UNIVERSITY

Summary

Sizing Agile Projects Using Function Points

https://www.youtube.com/watch?v=Xl_iuFSow6A

DUY TÂN UNIVERSITY

References

- Albrecht, A. J., & Gaffney, J. E. (1983). Software function, source lines of code, and development effort prediction: a software science validation. *IEEE transactions on software engineering*, (6), 639-648.

- Function Points Analysis Training Manual (110 Pages)- https://www.softwaremetrics.com/freemanual.htm

- **http://softwarecost.org/tools/COCOMO/**
- **http://softwarecost.org/tools/COCOMO/**